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go through the previous chats from start to end completely which are attached files and responses how detailed- <https://chatgpt.com/c/6a7f33eb-07f4-83ee-8229-383f952dcf93> , <https://chatgpt.com/c/6a8a1a24-1218-83e9-9f55-db7ec639b9a5> , <https://chatgpt.com/c/6a8a20f0-f3fc-83ee-891b-ba811975393f> ,

And continue further according to previous - Module 1,2,3 completed as above their links given and also attached files . Now continue with **Module 4: 75 REST/API questions.**

For your goal, I recommend turning this into a structured “**2-Year Java + Spring Boot Product Company Interview Bible**” with:

Module 1: 150 Core Java questions + answers

Module 2: 120 Spring/Spring Boot questions + internal working

Module 3: 80 JPA/Hibernate questions

Module 4: 75 REST/API questions

Module 5: 75 Microservices questions

Module 6: 50 Spring Security/JWT questions

Module 7: 50 SQL questions + queries

Module 8: 50 Multithreading questions

Module 9: 50 Kafka + Redis questions

Module 10: 50 Docker/AWS questions

Module 11: 100 DSA problems specifically for 2-year Java SDE interviews

Module 12: 50 LLD/system-design scenarios

Module 13: 100 project/resume follow-up questions

Module 14: Company-specific patterns for **Amazon, Google, Microsoft, Adobe, Uber, Walmart, Mastercard, Morgan Stanley, Cisco, Salesforce, Atlassian, Oracle, etc.**

Most importantly, for every question I'd structure it as:

Question → Short interview answer → Detailed explanation → Real project example → Follow-up question → Tricky interviewer counter-question → Best answer → Code/diagram where relevant

That format will train you to **answer follow-up questions instead of just memorizing definitions.**

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And continue further according to previous - Module 1,2,3,4 completed as above their links given and also attached files . Now continue with **Module 5: 75 Microservices questions** .

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chatgpt response: Yes. I've

Yes. I've reviewed the **attached conversation exports and the continuation material** to preserve the same progression, terminology, depth, and interview style.

One important limitation from the earlier chats: the `chatgpt.com/c/...` links themselves are not sufficient for me to retrieve private conversation contents, so I will **not claim that I read those URLs directly**. However, the attached DOCX exports contain the previous responses and continuation points, including Module 1–4 material, and I have used those as the source of continuity. For example, the REST export explicitly establishes Module 4 as 75 questions and says Module 5 should go deeper into distributed transactions/Saga design.

The original interview-bible structure is also preserved: **Question → Short interview answer → Detailed explanation → Real project example → Follow-up → Tricky counter-question → Best answer → Code/diagram where relevant**.

The previous material already introduced Microservices through **Monolith vs Microservices, API Gateway, Service Discovery, Load Balancing, Circuit Breaker and Distributed Transactions**, so I will **not simply repeat those definitions**.

□ Module 5 — Microservices

75 Interview Questions + Answers

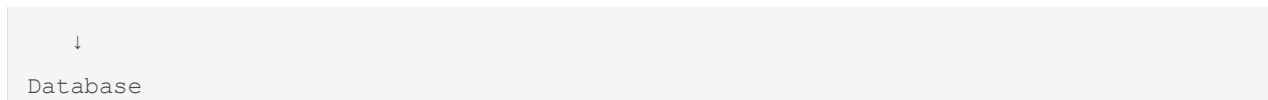
For 2-Year Java + Spring Boot Product-Company Interviews

How this module connects to Modules 1–4

The mental model we have built so far is:

text

```
Client
  ↓
REST API
  ↓
Controller
  ↓
Service
  ↓
@Transactional
  ↓
Spring Data JPA
  ↓
Hibernate
```



Now we introduce distributed architecture:



The key difference is that we now have **multiple independently running services**, network communication, failures, independent databases, eventual consistency, retries, messaging, observability and distributed transactions.

This is exactly the direction established in the earlier material: microservices provide independent deployment/scaling but introduce network failures, distributed transactions, observability, deployment and consistency challenges.

PART A — MICROSERVICES FUNDAMENTALS & ARCHITECTURE

Q1. What are Microservices? ☐☐☐

Short interview answer

Microservices is an architectural style where an application is divided into **small, independently deployable services**, each responsible for a specific business capability.

Each service can potentially have:

- its own codebase
- its own database
- its own deployment
- its own scaling
- its own technology choices

Detailed explanation

Instead of:

text

Monolith

User

Order

Payment

Inventory

Notification

↓

One application

↓

One deployment

↓

One database

we can have:

text

User Service

↓

User DB

Order Service

↓

Order DB

Payment Service

↓

Payment DB

Services communicate through:

text

REST

gRPC

Kafka

RabbitMQ

etc.

Real project example

For an e-commerce application:

```
text
User Service
Order Service
Payment Service
Inventory Service
Notification Service
```

The Order Service should not directly manipulate the Payment Service's database.

Follow-up

Why do companies use microservices?

Because they can independently:

- deploy services
- scale services
- isolate failures
- organize teams around business capabilities

Tricky counter-question

Are microservices always better than monoliths?

No.

Best answer

Microservices provide independent deployment and scaling, but they introduce distributed-system complexity such as network failures, service discovery, observability, distributed transactions and eventual consistency. I would choose microservices when the organizational and scalability requirements justify that complexity.

Q2. Monolith vs Microservices? ☐ ☐

Short answer

A monolith deploys the application as one unit, while microservices split it into independently deployable services.

Comparison

Monolith	Microservices
One deployable unit	Multiple deployable units

Monolith	Microservices
Often shared DB	Usually service-owned DB
Simple local communication	Network communication
Easier transactions	Distributed transactions are harder
Simple deployment	More deployment complexity
Easier initially	Better independent scaling
Failure can affect whole app	Failures can sometimes be isolated

Example

Monolith:

```

text
Order
Payment
Inventory
User
  ↓
One application

```

Microservices:

```

text
Order Service
Payment Service
Inventory Service
User Service

```

Tricky question

Why not make every class a microservice?

Because service boundaries should represent **business capabilities**, not arbitrary technical decomposition.

Q3. What is the biggest disadvantage of Microservices? ☐

Short answer

The biggest disadvantage is **distributed-system complexity**.

What changes?

In a monolith:

```
text
Order → Payment
```

may simply be:

```
java
paymentService.pay();
```

In microservices:

```
text
Order Service
    ↓ HTTP
Payment Service
```

Now you must consider:

- network timeout
- DNS
- service unavailable
- retries
- duplicate requests
- authentication
- serialization
- latency
- partial failure

Interview answer

Microservices move complexity from in-process communication to distributed communication.

That's a very strong statement to remember.

Q4. How do you identify Microservice boundaries? ☐

Short answer

Define services around **business capabilities** / **bounded contexts**, not technical layers.

Bad:

text

Controller Service
Repository Service
Database Service

Better:

text

Order Service
Payment Service
Inventory Service

Example

E-commerce:

text

Order
Payment
Inventory
Shipping
Notification

Each represents a business responsibility.

Follow-up

Should Order Service own payment data?

Generally, no. Payment data should be owned by Payment Service according to the chosen domain boundaries.

Tricky question

Can two services share the same database?

Technically yes, but it creates strong coupling and weakens service ownership.

Q5. What does "independent deployment" mean?

Short answer

A service can be changed and deployed without requiring all other services to be deployed together.

Example:

text

Order Service v3

Payment Service v7

Inventory Service v4

Order Service can be deployed independently.

Why important?

Suppose Payment Service has a bug.

You shouldn't need to redeploy:

text

User

Order

Inventory

Notification

just to fix Payment.

Follow-up

Does independent deployment mean no dependencies?

No.

Services still depend on APIs/events/contracts.

The goal is to minimize **tight coupling**.

Q6. What is bounded context?

Short answer

A bounded context is a boundary within which a particular domain model and terminology have a specific meaning.

Example

Order might mean:

text

Order Service:

Order = customer's purchase

while Payment might focus on:

```
text
```

```
Payment = financial transaction
```

Why useful?

It prevents one giant shared domain model across every service.

Tricky question

Is bounded context exactly the same as a microservice?

No.

A bounded context is a domain-design concept. A microservice can be an implementation of a bounded context, but the two are not strictly identical.

Q7. What is service autonomy?

Short answer

A service should be able to evolve, deploy and operate with minimal dependency on other services.

Example

Payment Service should own:

```
text
```

```
Payment logic
```

```
Payment state
```

```
Payment database
```

```
Payment APIs/events
```

rather than requiring Order Service to directly modify payment tables.

Strong answer

Service autonomy reduces coupling and allows independent development and deployment.

Q8. Why should each Microservice ideally own its database? ☐ ☐

Short answer

Database ownership prevents tight coupling and allows each service to evolve its data model independently.

Example

text

Order Service → order_db

Payment Service → payment_db

Inventory Service → inventory_db

Not:

text

Order ↘

Payment |— Shared DB

Inventory ↙

Why shared DB is dangerous?

Suppose Payment changes:

text

payment.status

Order Service may depend directly on that schema.

Now database schema changes become cross-service changes.

Follow-up

How does Order get payment information then?

Through:

text

REST

or

events

Q9. What is the Database-per-Service pattern? ★

Short answer

Each microservice owns its persistent data store.

text

Order Service

↓

```
Order DB

Payment Service
  ↓
Payment DB
```

Benefit

Service autonomy.

Cost

Cross-service queries become difficult.

You cannot simply do:

```
sql
SELECT *
FROM orders
JOIN payments ...
```

across independent databases.

Solution

Use:

- APIs
- events
- materialized views
- CQRS/read models
- data synchronization

Q10. Why is a shared database considered an anti-pattern in many Microservice designs?

Short answer

Because it creates tight coupling between services.

Example

```
text
Order Service
  ↓
```

```
DB
↑
Payment Service
```

Now both services know:

- tables
- columns
- constraints
- indexes
- relationships

This makes independent deployment harder.

Tricky question

Can shared DB ever be acceptable?

Yes, especially during migration or where organizational/technical constraints justify it.

The important point is to understand the trade-off.

Q11. What is loose coupling? ☐

Short answer

Loose coupling means services depend on stable contracts rather than internal implementation details.

Bad:

```
text
Order Service
↓
Payment DB
```

Better:

```
text
Order Service
↓ API/Event
Payment Service
```

Example

Payment Service can internally change:

text

MySQL → PostgreSQL

without requiring Order Service to change, assuming the API contract remains compatible.

Q12. What is high cohesion?

Short answer

A service should contain functionality that strongly belongs together.

For example:

text

```
Payment Service
├─ Payment creation
├─ Payment status
├─ Refund
├─ Payment validation
└─ Payment integration
```

This is better than scattering payment functionality across unrelated services.

Interview phrase

High cohesion within a service and loose coupling between services.

Q13. What is synchronous communication? ☐

Short answer

The caller waits for the downstream service to return a response.

text

```
Order Service
    ↓ HTTP
Payment Service
    ↓
Response
    ↓
Order Service continues
```

Example

```
java
PaymentResponse response =
    paymentClient.pay(request);
```

Advantages

- simple request/response model
- immediate result
- easy to understand

Disadvantages

- network latency
- timeout
- dependency availability
- cascading failures

Q14. What is asynchronous communication?

Short answer

The sender sends a message/event and does not necessarily wait for the consumer to complete processing.

```
text
Order Service
    ↓
Event
    ↓
Kafka
    ↓
Payment Service
```

Example

```
text
OrderCreated
```

can be consumed by:

```
text
Payment
Inventory
```



```
Notification
Analytics
```

Benefit

Reduced temporal coupling.

Q15. REST vs Messaging in Microservices? ☐ ☐

REST

```
text
A → request → B
A ← response ← B
```

Good when:

- immediate response required
- synchronous workflow
- query operation

Messaging

```
text
A → Event → Broker → B
```

Good when:

- asynchronous processing
- event-driven workflows
- decoupling
- buffering
- replay

Tricky question

Is Kafka always better than REST?

No.

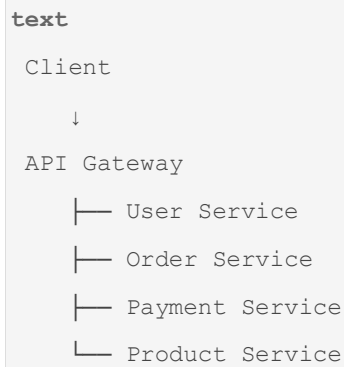
The previous material explicitly warned against saying Kafka is automatically better than REST.

PART B — API GATEWAY, SERVICE DISCOVERY & COMMUNICATION

Q16. What is an API Gateway? □ □ □

Short answer

An API Gateway is an entry point between clients and backend services.



Responsibilities

Potentially:

- routing
- authentication
- rate limiting
- TLS termination
- logging
- metrics
- request transformation

The previous REST module already established this architecture.

Tricky question

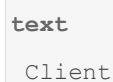
Should business logic be placed in Gateway?

Generally no.

Gateway should handle cross-cutting/API-edge concerns; business logic belongs in domain services.

Q17. Why use an API Gateway?

Benefits



```
↓  
Gateway  
↓  
Services
```

Instead of clients knowing:

```
text  
user-service  
order-service  
payment-service  
inventory-service
```

the client knows one entry point.

Other benefits

- centralized authentication
- rate limiting
- routing
- monitoring
- request aggregation

Follow-up

What's the downside?

Gateway can become:

```
text  
Single bottleneck  
Single failure point  
Business-logic dumping ground
```

So it must be highly available and carefully designed.

Q18. What is service discovery? ☐ ☐

Short answer

Service discovery allows services to dynamically locate other service instances.

Instead of:

```
text
```

```
http://10.20.10.3:8080
```

we can discover:

```
text
```

```
payment-service
```

and obtain available instances.

The earlier material specifically introduced this problem of dynamic service instances.

Why needed?

Instances can:

- start
- stop
- scale
- move
- change IP

Hardcoding IP addresses is therefore fragile.

Q19. Client-side vs Server-side service discovery?

Client-side

```
text
```

```
Client
```

```
↓
```

```
Discovery Service
```

```
↓
```

```
choose instance
```

```
↓
```

```
Service
```

Client performs discovery/load balancing.

Server-side

```
text
```

```
Client
```

```
↓
```

```
Load Balancer
```

```
↓
```

```
Discovery
```

```
↓
```

```
Service
```

Infrastructure handles routing.

Interview answer

The distinction is mainly who is responsible for discovering/selecting the service instance: the client or an intermediary.

Q20. What is service registry?

Short answer

A service registry maintains information about available service instances.

Conceptually:

```
text
```

```
Service Registry
```

```
payment-service
```

```
├─ instance-1
```

```
├─ instance-2
```

```
└─ instance-3
```

```
order-service
```

```
├─ instance-1
```

```
└─ instance-2
```

Services register themselves and consumers can discover them.

Q21. What happens when a service instance crashes?

Strong answer

The system should detect that the instance is unhealthy and stop routing traffic to it.

Typical mechanism:

```
text
Instance
  ↓
Health Check
  ↓
Registry / Load Balancer
```

If unhealthy:

```
text
Remove / mark unavailable
```

Follow-up

Why health checks matter?

Because "process is running" does not necessarily mean "service is healthy."

Q22. What is load balancing? ☐

Short answer

Load balancing distributes requests across multiple service instances.

```
text
Load Balancer
  /   |   \
  ↓   ↓   ↓
App1 App2 App3
```

Benefits include:

- scalability
- availability
- traffic distribution

This was explicitly identified as an interview topic in the previous material.

Q23. What is horizontal scaling?

Short answer

Add more service instances rather than making one instance bigger.

text

Before:

Order Service × 1

After:

Order Service × 5

Traffic is distributed across instances.

Vertical scaling

text

2 CPU → 8 CPU

Horizontal

text

1 instance → 5 instances

Microservices commonly benefit from horizontal scaling.

Q24. What is client-side load balancing?

The caller knows multiple service instances and chooses one.

text

Order Service

↓

Instance List

├─ Payment-1

├─ Payment-2

└─ Payment-3

It can use algorithms such as:

- round robin
- random
- weighted
- least connections

Q25. What is server-side load balancing?

The client sends requests to a load balancer:

```
text
Order
  ↓
Load Balancer
  ↓
Payment-1 / Payment-2 / Payment-3
```

The client doesn't need to know individual instances.

Q26. What is an inter-service contract? ☐

Short answer

A contract defines how one service communicates with another.

For REST:

```
text
POST /payments
```

may define:

```
json
{
  "orderId": 1001,
  "amount": 500
}
```

and response:

```
json
{
  "paymentId": 2001,
  "status": "SUCCESS"
}
```

Why important?

Changing contracts carelessly can break consumers.

This connects directly to the backward-compatibility concerns established in Module 4.

Q27. How do you maintain backward compatibility between Microservices?

Techniques

Avoid:

```
text
Rename field
Delete field
Change meaning
```

without compatibility planning.

Prefer:

```
text
Old field remains
+
new field added
```

or:

```
text
API v1
API v2
```

Strong answer

In distributed systems, contract changes must be treated as compatibility problems because producers and consumers may be deployed independently.

Q28. What is API versioning in Microservices?

Examples:

```
text
/api/v1/orders
/api/v2/orders
```

or header-based versioning.

Why?

Consumers may not upgrade simultaneously.

Tricky question

Should every service have a new version for every change?

No.

Backward-compatible changes often don't require a new major API version.

Q29. What is service-to-service authentication?

Short answer

Authentication between backend services ensures that one service can prove its identity to another.

Example:

```
text
Order Service
    ↓ token
Payment Service
```

Possible mechanisms:

- OAuth2 client credentials
- mTLS
- signed service tokens
- platform identity

Important

Don't rely merely on:

```
text
"it's internal network, so it's trusted."
```

Q30. REST vs gRPC for Microservices?

REST

Advantages:

- simple
- widely supported
- browser-friendly
- JSON ecosystem

gRPC

Advantages:

- efficient binary serialization
- strongly typed contracts
- good internal service communication
- streaming support

Interview answer

I would choose based on communication requirements rather than claiming one is universally better.

PART C — RESILIENCE, TIMEOUTS, RETRIES & FAILURE

This section is extremely important because this is where a basic REST candidate becomes a **distributed-systems candidate**.

Q31. Why are network calls fundamentally different from method calls?

□ □ □

Consider:

```
java
paymentService.pay();
```

In-process call:

```
text
Application memory
```

Microservice call:

```
text
Order
↓
Network
↓
Payment
```

The network can:

- delay

- drop packets
- timeout
- duplicate requests
- disconnect
- return errors

Best interview statement

A remote call should always be treated as unreliable compared with an in-process method call.

Q32. What is a timeout? ☐ ☐

Short answer

A timeout limits how long a caller waits for a downstream operation.

Example:

```
text
Order Service
    ↓
Payment Service
    ↓
No response
    ↓
5-second timeout
```

Without timeout:

```
text
Request waits indefinitely
```

which can consume:

- threads
- connections
- memory

Rule

Every remote call should have an appropriate timeout.

Q33. Connection timeout vs Read timeout?

Connection timeout

Time allowed to establish a connection.

```
text
Client → establish connection
```

Read timeout

Time allowed waiting for data after connection.

```
text
Request sent
  ↓
waiting for response
```

Interview trap

A connection can succeed while the response still takes too long.

Therefore these are separate concerns.

Q34. What is retry? ☐

Retry means attempting an operation again after a failure.

```
text
Request
  ↓
Failure
  ↓
Retry
  ↓
Success
```

Useful for transient failures.

Examples:

- temporary network failure
- transient service unavailable
- temporary infrastructure issue

Dangerous

Retrying non-idempotent operations blindly.

Q35. Why can retries create duplicate transactions? ☐ ☐ ☐

Suppose:

```
text
Order Service → Payment
```

Payment succeeds:

```
text
₹100 charged
```

But response is lost.

Order Service thinks:

```
text
timeout
```

and retries.

Payment receives:

```
text
same payment
```

again.

Potentially:

```
text
₹100
+
₹100
=
₹200 charged
```

Solution

Use:

```
text
Idempotency Key
```

The REST module already established this as essential for payment APIs.

Q36. What is exponential backoff? ☐

Instead of:

```
text
retry
retry
retry
```

use:

```
text
100ms
200ms
400ms
800ms
```

The previous REST material explicitly established this progression.

Why?

It reduces pressure on an unhealthy dependency.

Q37. What is jitter? ☐

If thousands of clients all retry at exactly:

```
text
1 second
```

they may create another traffic spike.

Jitter introduces randomness:

```
text
800ms + random delay
```

Strong answer

Exponential backoff controls retry frequency, while jitter prevents synchronized retries.

Q38. What is a retry storm? ☐ ☐

Suppose:

```
text
100 requests
x
3 retries
```

Potential downstream requests:

```
text
100 × 4 = 400
```

The original failure becomes worse.

The previous REST module explicitly calls this a **retry storm**.

Prevention

```
text
Timeout
+
Bounded retry
+
Backoff
+
Jitter
+
Circuit breaker
+
Idempotency
```

Q39. What is a Circuit Breaker? ☐☐☐

Short answer

A circuit breaker prevents repeated calls to an unhealthy dependency.

States:

```
text
CLOSED
  ↓ failures
OPEN
  ↓ wait
HALF_OPEN
```



```
↓ test  
CLOSED / OPEN
```

The previous material establishes exactly this model.

CLOSED

Requests flow normally.

OPEN

Requests fail fast.

HALF_OPEN

Limited test requests determine whether the dependency recovered.

Q40. What is cascading failure? ☐ ☐

Suppose:

```
text  
Order  
↓  
Payment  
↓  
Fraud  
↓  
External Bank
```

Bank becomes slow.

Then:

```
text  
Fraud waits  
↓  
Payment waits  
↓  
Order waits  
↓  
threads exhausted
```

Eventually the entire system becomes unhealthy.

Prevention

- timeout
- circuit breaker
- bulkheads
- bounded retries
- async processing

Q41. What is the Bulkhead pattern?

Short answer

Bulkhead isolates resources so that failure in one dependency doesn't consume resources needed by others.

Example:

```
text
Service
├─ Payment thread pool
├─ Inventory thread pool
└─ Notification thread pool
```

If Payment becomes slow:

```
text
Payment pool exhausted
```

but Inventory may continue working.

Analogy

Like compartments in a ship.

One compartment flooding shouldn't sink the whole ship.

Q42. What is graceful degradation?

Short answer

When a dependency fails, the system provides reduced functionality rather than complete failure.

Example:

```
text
Recommendation Service DOWN
```

Instead of:

text

Product API → 500

return:

text

Product API → product data

+

no recommendations

Strong answer

Non-critical dependencies should often fail independently without taking down the core business operation.

Q43. What is a fallback?

A fallback is an alternative response/behavior when a dependency fails.

Example:

text

Recommendation Service

↓

timeout

↓

fallback

↓

return products without recommendations

Tricky question

Should every failure have a fallback?

No.

A fallback must preserve business correctness.

For example, silently assuming:

text

Payment succeeded

when Payment Service is unavailable would be dangerous.

Q44. What is fault tolerance?

Short answer

Fault tolerance means the system continues providing acceptable behavior despite failures in some components.

Techniques:

text

- Replication
- Load balancing
- Timeout
- Retry
- Circuit breaker
- Bulkhead
- Fallback
- Queueing

Q45. What is resilience vs availability?

Availability

Is the system operational?

Resilience

How well does the system handle and recover from failures?

A service can have high availability but poor resilience if a small dependency failure causes a major outage.

PART D — DISTRIBUTED TRANSACTIONS, SAGA & CONSISTENCY

This is one of the **highest-value sections for a 2-year product-company interview**.

The previous REST module intentionally ended by saying that external Payment + Order transaction crosses into distributed transaction/Saga design, which is now the focus of Module 5.

Q46. What is a distributed transaction? ☐ ☐ ☐

Suppose:

text

```
Order Service
    ↓
Payment Service
    ↓
Inventory Service
```

All operations need to succeed.

In a monolith:

text

```
One DB transaction
```

might handle this.

In microservices:

text

```
Order DB
Payment DB
Inventory DB
```

are separate.

Therefore one normal local DB transaction cannot atomically cover all of them.

The previous material establishes exactly this problem.

Q47. Why is `@Transactional` not enough for Microservices? ☐☐☐

Suppose:

java

```
@Transactional
public void createOrder() {
    saveOrder();
    paymentClient.pay();
}
```

The transaction normally manages your local transactional resource/database.

It does **not automatically make the external Payment Service part of the same local transaction.**

If:

text

```
DB commit succeeds  
  
but  
  
Payment succeeds/fails independently
```

you can have inconsistent state.

Strong answer

@Transactional gives local transaction semantics; it does not automatically provide atomicity across independent microservices.

Q48. What is Saga Pattern? □ □ □

Short answer

Saga manages a distributed business transaction as a sequence of local transactions with compensating actions.

Example:

text

```
Create Order  
    ↓  
Reserve Inventory  
    ↓  
Process Payment  
    ↓  
Confirm Order
```

If Payment fails:

text

```
Cancel Order  
  
+  
  
Release Inventory
```

The earlier material explicitly identified Saga, eventual consistency and compensating actions as approaches to distributed transactions.

Q49. What is a compensating transaction?

A compensating transaction semantically reverses a previous successful business operation.

Example:

text

Reserve Inventory

If later Payment fails:

text

Release Inventory

It isn't necessarily a database rollback.

That's the key distinction.

Database rollback

text

UNDO DB transaction

Compensation

text

Perform another business operation

Q50. Choreography vs Orchestration in Saga? ☐ ☐

Choreography

Services react to events.

text

OrderCreated

↓

Inventory consumes

↓

InventoryReserved

↓

Payment consumes

↓

PaymentCompleted

No central coordinator.

Advantage

Loose coupling.

Disadvantage

Workflow can become difficult to understand.

Orchestration

A central orchestrator controls the workflow.

```
text
Saga Orchestrator
    ↓
Create Order
    ↓
Reserve Inventory
    ↓
Process Payment
    ↓
Confirm Order
```

Advantage

Workflow easier to understand.

Disadvantage

Orchestrator can become complex or overly central.

Q51. What is eventual consistency? ☐ ☐ ☐

Short answer

Eventual consistency means distributed replicas/services may temporarily have different states, but the system converges toward a consistent state.

Example:

```
text
Order = CREATED
```

Payment may initially show:

```
text
PENDING
```

Later:


```
text
```

```
Payment = SUCCESS
```

```
Order = CONFIRMED
```

Important

Not every system requires immediate strong consistency.

Q52. Strong consistency vs Eventual consistency?

Strong consistency

After successful write, reads observe the latest state according to the consistency model.

Eventual consistency

Temporary inconsistency is acceptable.

Example

Bank balance:

```
text
```

```
Strong consistency often important
```

Product recommendation count:

```
text
```

```
Eventual consistency may be acceptable
```

Interview answer

The choice depends on business correctness requirements rather than simply saying strong consistency is always better.

Q53. What is a distributed lock?

A distributed lock coordinates access to a shared logical resource across multiple application instances.

Example:

```
text
```

```
Order instances:
```

```
App1
```

```
App2
App3
```

All attempt:

```
text
process order 100
```

A distributed lock can ensure only one instance owns the operation at a time.

But beware

Distributed locks introduce their own problems:

- lease expiry
- network partitions
- clock assumptions
- lock ownership
- failure recovery

Don't use them automatically when an idempotency key or database constraint can solve the problem more safely.

Q54. How do you prevent duplicate order creation?

This is a very strong interview scenario.

Use:

```
text
Idempotency-Key
+
Database unique constraint
+
Atomic state transition
```

Example:

```
text
POST /orders
Idempotency-Key: abc123
```

Database:

text

```
UNIQUE(customer_id, idempotency_key)
```

The previous Spring material also established that application-level `exists` checks alone are unsafe under concurrency and that a database unique constraint is the final protection.

Q55. Why is `check-then-insert` unsafe?

Bad:

java

```
if (!exists(key)) {  
    insert(key);  
}
```

Two requests:

text

```
A → exists? NO  
B → exists? NO  
  
A → INSERT  
B → INSERT
```

Correct approach

Database constraint:

sql

```
UNIQUE(idempotency_key)
```

Then one insert succeeds and the other fails safely.

Q56. What happens if Payment succeeds but Order DB update fails? ☐☐☐

This is a classic interview question.

Suppose:

text

```
Order = CREATED  
  
Payment = SUCCESS
```

```
Order update = FAILED
```

Now:

```
text
Money taken
Order still says CREATED
```

Strong answer

Don't claim that `@Transactional` solves it.

Instead:

```
text
Explicit state machine
+
Reliable events/outbox
+
Retry/reconciliation
+
Idempotent payment operation
```

Potential state:

```
text
PAYMENT_PENDING
PAYMENT_SUCCESS
PAYMENT_FAILED
RECONCILIATION_REQUIRED
```

The earlier Spring module explicitly identified this exact distributed-consistency problem and recommended reliable workflow/outbox/reconciliation thinking.

Q57. What is the Outbox Pattern? ☐ ☐ ☐

Problem

Suppose:

```
text
DB transaction
```

```
+  
publish Kafka event
```

You do:

```
text  
saveOrder()  
publish(OrderCreated)
```

DB commit succeeds but Kafka publishing fails.

Now:

```
text  
Order exists  
Event doesn't
```

Outbox solution

Within the same local DB transaction:

```
text  
orders table  
+  
outbox table  
  
text  
@Transactional  
↓  
Create Order  
+  
Create Outbox Event  
↓  
COMMIT
```

Then a publisher reads the outbox:

```
text  
Outbox  
↓  
Kafka  
↓  
Consumers
```

Why powerful?

The order and event record are committed atomically in the same local DB transaction.

Q58. What is an Outbox Event?

Example:

```
json
{
  "eventId": "evt-123",
  "type": "OrderCreated",
  "aggregateId": "order-1001",
  "payload": {}
}
```

Stored in:

```
text
outbox_event
```

Then asynchronously published.

Follow-up

Can Kafka publishing still fail?

Yes.

The publisher retries.

Therefore consumers should also be designed for duplicate delivery.

Q59. What is Idempotent Consumer? ☐ ☐

Suppose Kafka delivers:

```
text
OrderCreated
```

twice.

Consumer must ensure processing twice doesn't cause two business effects.

Example:

```
text
eventId = evt-123
```

Consumer stores processed IDs:

```
text
processed_event
```

If:

```
text
evt-123 already processed
```

ignore duplicate.

Strong answer

At-least-once delivery often means consumers must be idempotent.

Q60. At-most-once vs At-least-once vs Exactly-once?

At-most-once

Message processed:

```
text
0 or 1 time
```

Could be lost.

At-least-once

Message processed:

```
text
1 or more times
```

Could be duplicated.

Exactly-once

Intended effect occurs exactly once under the system's defined semantics.

Interview trap

Don't casually say:

"Kafka gives exactly-once, so duplicates don't matter."

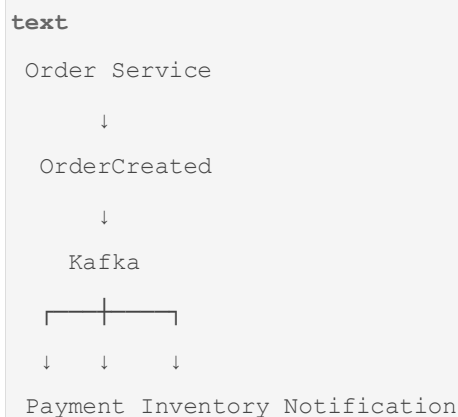
Exactly-once semantics are nuanced and do not magically make arbitrary external side effects exactly-once.

PART E — EVENT-DRIVEN MICROSERVICES, OBSERVABILITY & PRODUCTION SCENARIOS

Q61. What is Event-Driven Architecture? ☐ ☐

Short answer

Services communicate by publishing and consuming events.



Advantages

- loose coupling
- asynchronous processing
- scalability
- buffering
- event replay

Disadvantages

- eventual consistency
- debugging complexity
- duplicate events
- ordering concerns
- schema evolution

Q62. Event vs Command? ☐

Command

Tells another component to perform an action.

```
text
ProcessPayment
```

Event

States something already happened.

```
text
PaymentCompleted
```

Memory trick

```
text
Command = "Do this"

Event = "This happened"
```

Q63. What is event ordering?

Suppose:

```
text
OrderCreated
OrderCancelled
```

If consumers receive:

```
text
OrderCancelled
OrderCreated
```

incorrect state can occur.

Therefore some workflows require ordering guarantees.

Solution depends on broker/design

For Kafka, partitioning strategy can help maintain order for messages within the same partition.

Interview trap

Kafka does not provide global ordering across all partitions.

Q64. What is a Dead Letter Queue (DLQ)?

A DLQ stores messages that repeatedly fail processing.

```
text
Kafka
↓
Consumer
↓
Processing failure
↓
Retry
↓
Retry
↓
DLQ
```

Why?

Without DLQ:

```
text
bad message
↓
retry forever
↓
consumer blocked
```

DLQ helps

- isolate poison messages
- investigate failures
- retry later

Q65. What is a poison message?

A message that repeatedly fails processing.

Example:

```
json
{
  "quantity": "INVALID"
}
```

Consumer keeps failing.

Correct strategy

```
text
Bounded retries
+
DLQ
+
alerting
+
manual/automated remediation
```

Q66. What is schema evolution in event-driven systems?

Suppose event version 1:

```
json
{
  "orderId": 100
}
```

Later:

```
json
{
  "orderId": 100,
  "customerId": 50
}
```

Consumers must continue functioning.

Best practices

- backward-compatible changes
- additive fields
- schema versioning where needed
- avoid changing field meaning

Q67. What is observability in Microservices? ☐ ☐ ☐

With 20 services:

text

Client

↓

Gateway

↓

Order

↓

Payment

↓

Bank

If latency becomes 5 seconds, logs from one service aren't enough.

Observability typically includes:

text

Logs

Metrics

Traces

Previous modules already established this mindset for production API investigation.

Q68. What is distributed tracing? ☐ ☐

A trace follows one request across multiple services.

Example:

text

traceId = abc123

Gateway

↓ 10ms

Order

↓ 40ms

Payment

↓ 2.5 sec

Bank

Now we know:

```
text
```

```
Bank = latency source
```

Without tracing

You may only know:

```
text
```

```
Order API = 2.6 seconds
```

but not why.

Q69. Why is correlation ID important?

A correlation ID lets you connect logs belonging to the same request/workflow.

Example:

```
text
```

```
X-Correlation-ID: abc123
```

Logs:

```
text
```

```
Order Service → abc123
```

```
Payment Service → abc123
```

```
Inventory Service → abc123
```

This makes distributed debugging much easier.

Q70. Design a high-throughput Order API using Microservices. □□□□

Suppose:

```
text
```

```
50,000 requests/sec
```

Architecture:

```
text
```

```
Client
```

```
↓
```

```
CDN / Load Balancer
```

```
↓
```

```
API Gateway
  ↓
Order Service × N
  ↓
Order DB
  ↓
Outbox
  ↓
Kafka
  ├── Payment
  ├── Inventory
  └── Notification
```

Use:

text

```
Horizontal scaling
Caching where appropriate
Database indexes
Connection pooling
Async processing
Partitioning
Observability
Rate limiting
```

Interview trap

Don't immediately say:

Add Redis and Kafka.

First ask:

- What is the traffic pattern?
- Which operations are synchronous?
- What consistency is required?
- What is the bottleneck?
- What data is cacheable?

Q71. Design an Order → Payment → Inventory workflow. □ □ □ □

A strong design:

text

```
Client
  ↓
Order Service
  ↓
Create Order = PENDING
  ↓
Outbox
  ↓
Kafka
  ↓
Payment
  ↓
PaymentSucceeded
  ↓
Inventory
  ↓
InventoryReserved
  ↓
Order = CONFIRMED
```

If Payment fails:

text

```
PaymentFailed
  ↓
Order = PAYMENT_FAILED
```

If Inventory fails:

text

```
Release/Refund Payment
  ↓
Order = FAILED
```

This is essentially Saga-style distributed business workflow.

What interviewer wants

They want you to mention:

- local transactions

- events
- idempotency
- compensation
- eventual consistency
- retries
- failure handling

Q72. What happens if one Microservice is down?

Don't simply answer:

API returns 500.

Analyze dependency importance.

Critical dependency

```
text
Payment DOWN
```

Maybe:

```
text
Order = PAYMENT_PENDING
```

rather than pretending payment succeeded.

Non-critical dependency

```
text
Recommendation DOWN
```

Product API may still return:

```
text
products
+
recommendations unavailable
```

Strong answer

Failure behavior should be driven by business criticality, not merely technical availability.

Q73. Your Payment Service becomes slow. What would you do? ☐☐☐☐

Don't say:

Increase timeout.

Use a systematic approach.

```
text
Order
↓
Payment
```

Implement:

```
text
Timeout
↓
Bounded Retry
↓
Exponential Backoff
↓
Jitter
↓
Circuit Breaker
```

If payment is non-immediate:

```
text
Async workflow
+
Payment Pending state
```

Also inspect:

- Payment latency
- connection pool
- thread pool
- database
- external payment provider
- error rate
- p95/p99

This follows the same systematic production-investigation philosophy established in Module 4.

Q74. Production incident: Order API latency increases from 200 ms to 5 seconds. What do you do? ☐☐☐☐

This is the **Module 5 version of the production-debugging question**.

Given:

text

Normal:

200 ms

Current:

5 sec

DB CPU:

85%

Connection pool:

100%

External Payment:

8 sec

Step 1 — Confirm scope

Check:

text

p50

p95

p99

error rate

throughput

Step 2 — Distributed tracing

Determine:

text

Gateway = 10 ms

Order = 30 ms

DB = 2.5 sec

Payment = 1.8 sec

Step 3 — Database

Check:

```
text
N+1
slow queries
indexes
execution plan
locks
connection pool
```

Step 4 — Payment

Check:

```
text
timeout
retry count
dependency latency
circuit state
```

Step 5 — Check retry storm

Suppose:

```
text
100 requests
x
3 retries
```

Now downstream may receive:

```
text
400 requests
```

Step 6 — Mitigate

Depending on evidence:

```
text
Reduce/bound retries
Enable circuit breaker
Fix N+1
Restore cache
```

```
Scale service
Protect DB
```

The earlier Spring production scenario used the same reasoning approach: inspect traces, query counts, DB connections, cache hit rate and external dependency latency rather than blindly changing resources.

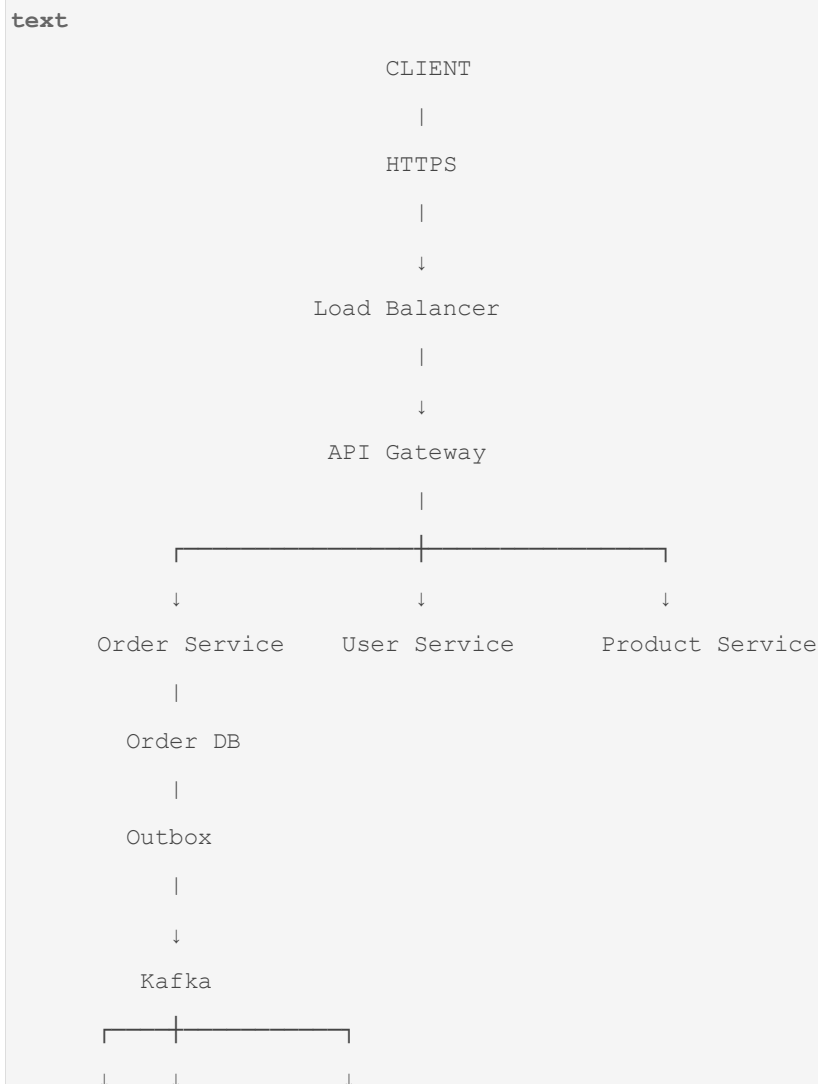
Q75. □□□□ Final Master Question — Design a Production-Grade Microservices E-Commerce System

This is the **most important question of Module 5**.

Suppose interviewer says:

"Design an Order system using microservices. Explain the architecture, communication, database strategy, transactions, failures, scalability and observability."

Your answer should look like this:



Payment	Inventory	Notification
Service	Service	Service
Payment DB	Inventory DB	Notification

Step 1 — API Layer

```
text
POST /api/v1/orders
Authorization: Bearer ...
Idempotency-Key: abc123
```

API Gateway handles:

```
text
Authentication
Routing
Rate limiting
TLS
Observability
```

The REST module already established this edge architecture.

Step 2 — Order Service

Validate:

```
text
customer
products
quantity
business rules
```

Then:

```
text
Order = PENDING
```

Step 3 — Database transaction

Use local transaction:

```
text
@Transactional
```

```
↓
Create Order
↓
Create Order Items
↓
Create Outbox Event
↓
COMMIT
```

Important:

Do not keep the DB transaction open while waiting for Payment Service.

The earlier modules explicitly warned against putting slow external payment calls inside the DB transaction.

Step 4 — Publish event

```
text
Outbox
↓
OrderCreated
↓
Kafka
```

Consumers:

```
text
Payment
Inventory
Notification
```

Step 5 — Payment

```
text
OrderCreated
↓
Payment Service
↓
Process payment
```

Use:

text

Idempotency
Timeout
Bounded retry
Backoff
Jitter
Circuit breaker

Step 6 — Inventory

text

PaymentSucceeded
↓
Inventory Service
↓
Reserve stock

If inventory fails:

text

Compensate payment
+
Cancel/reject order

Step 7 — Final Order state

Possible state machine:

text

PENDING
↓
PAYMENT_PENDING
↓
PAYMENT_SUCCESS
↓
INVENTORY_RESERVED
↓
CONFIRMED

Failure:

text

```
PAYMENT_FAILED  
INVENTORY_FAILED  
CANCELLED
```

Step 8 — Database ownership

text

```
Order → Order DB  
  
Payment → Payment DB  
  
Inventory → Inventory DB  
  
User → User DB
```

Don't allow:

text

```
Payment Service → Order DB directly
```

Step 9 — Idempotency

For:

text

```
POST /orders
```

use:

text

```
Idempotency-Key  
+  
Unique DB constraint
```

For events:

text

```
eventId  
+  
processed-event table
```

This protects against duplicate requests and duplicate event delivery.

Step 10 — Resilience

For synchronous calls:

```
text
Timeout
+
Bounded Retry
+
Exponential Backoff
+
Jitter
+
Circuit Breaker
```

For asynchronous workflows:

```
text
Kafka
+
Retry
+
DLQ
+
Idempotent Consumer
```

Step 11 — Observability

Every request should be traceable:

```
text
traceId
correlationId
orderId
```

Monitor:

```
text
Latency
p50
p95
p99
```

```
Error rate
Throughput
CPU
Memory
GC
DB latency
Kafka lag
Downstream latency
Circuit-breaker state
```

Step 12 — Scaling

If Order Service receives high traffic:

```
text
Order Service × 10
```

behind:

```
text
Load Balancer
```

Database can be optimized using:

```
text
Indexes
Connection pooling
Read replicas where appropriate
Caching where appropriate
Partitioning where justified
```

Kafka can provide:

```
text
Asynchronous processing
Buffering
Parallelism
Decoupling
```

☐ Module 5 — 75 Questions Master Recall List

Part A — Fundamentals

1. What are Microservices?
2. Monolith vs Microservices?
3. Biggest disadvantage of Microservices?
4. How do you identify service boundaries?
5. What is independent deployment?
6. What is bounded context?
7. What is service autonomy?
8. Why should services own their databases?
9. What is Database-per-Service?
10. Why is shared DB coupling?
11. What is loose coupling?
12. What is high cohesion?
13. What is synchronous communication?
14. What is asynchronous communication?
15. REST vs messaging?

Part B — Communication & Infrastructure

1. What is API Gateway?
2. Why use API Gateway?
3. What is Service Discovery?
4. Client-side vs server-side discovery?
5. What is Service Registry?
6. What happens when a service crashes?
7. What is Load Balancing?
8. What is Horizontal Scaling?
9. What is Client-side Load Balancing?
10. What is Server-side Load Balancing?
11. What is an inter-service contract?
12. How do you maintain backward compatibility?
13. What is API versioning?
14. What is service-to-service authentication?
15. REST vs gRPC?

Part C — Resilience

1. Why are remote calls different from method calls?
2. What is timeout?

3. Connection vs Read timeout?
4. What is retry?
5. Why do retries create duplicates?
6. What is exponential backoff?
7. What is jitter?
8. What is retry storm?
9. What is Circuit Breaker?
10. What is cascading failure?
11. What is Bulkhead?
12. What is graceful degradation?
13. What is fallback?
14. What is fault tolerance?
15. Resilience vs availability?

Part D — Distributed Transactions

1. What is distributed transaction?
2. Why isn't `@Transactional` enough?
3. What is Saga?
4. What is compensating transaction?
5. Choreography vs Orchestration?
6. What is eventual consistency?
7. Strong vs eventual consistency?
8. What is distributed locking?
9. How do you prevent duplicate orders?
10. Why is check-then-insert unsafe?
11. Payment succeeds but Order update fails?
12. What is Outbox Pattern?
13. What is Outbox Event?
14. What is Idempotent Consumer?
15. At-most-once vs At-least-once vs Exactly-once?

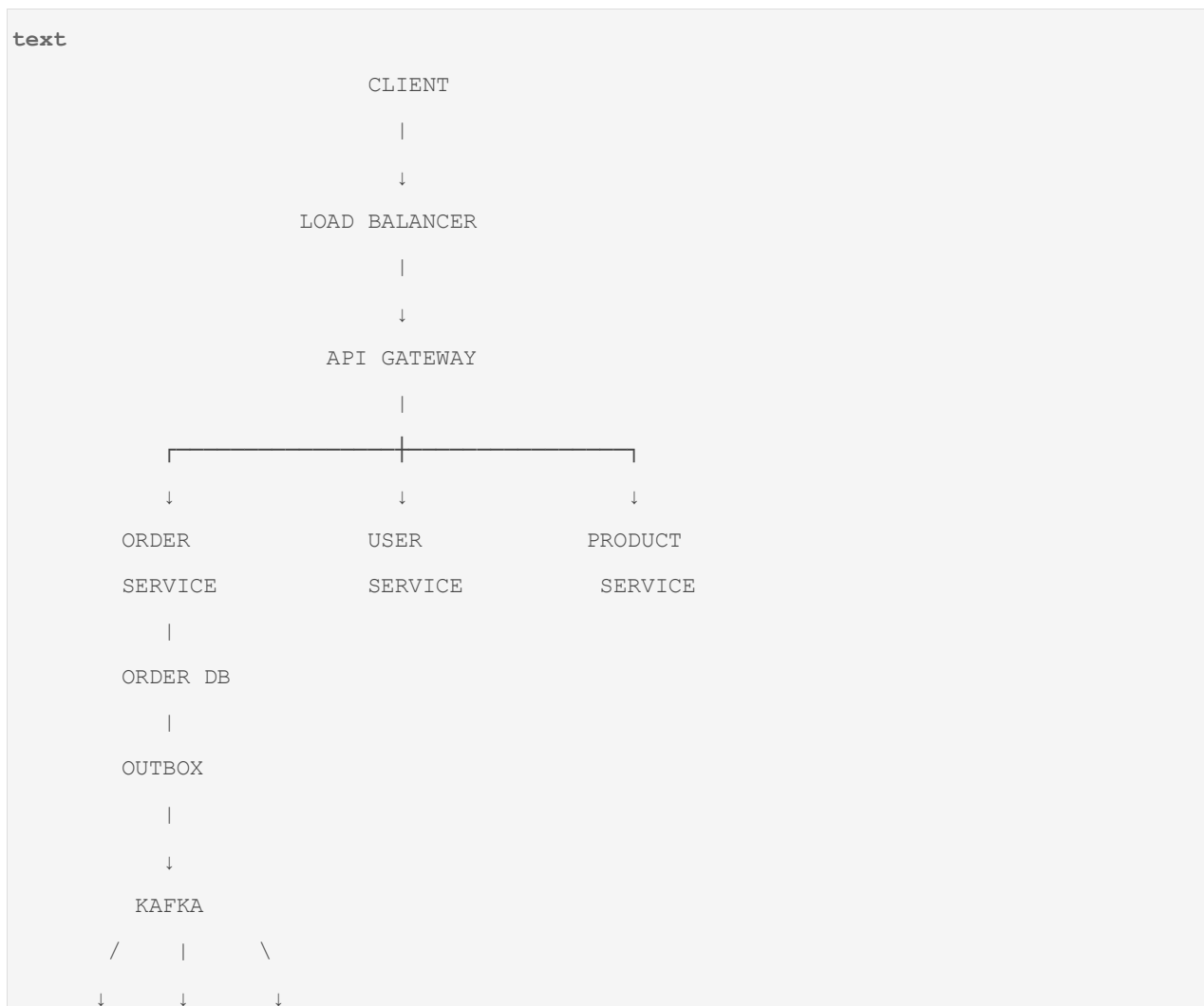
Part E — Production & Architecture

1. What is Event-Driven Architecture?
2. Event vs Command?
3. What is Event Ordering?
4. What is DLQ?

5. What is a Poison Message?
6. What is Schema Evolution?
7. What is Observability?
8. What is Distributed Tracing?
9. Why Correlation ID?
10. Design high-throughput Order API.
11. Design Order → Payment → Inventory.
12. What happens if a Microservice is down?
13. Payment Service becomes slow — what do you do?
14. Production API latency rises from 200 ms to 5 seconds — investigate.
15. **Design a complete production-grade Microservices E-commerce system.**

□ What you should be able to draw in an interview

By the end of Module 5, you should be able to draw this **without memorizing it**:

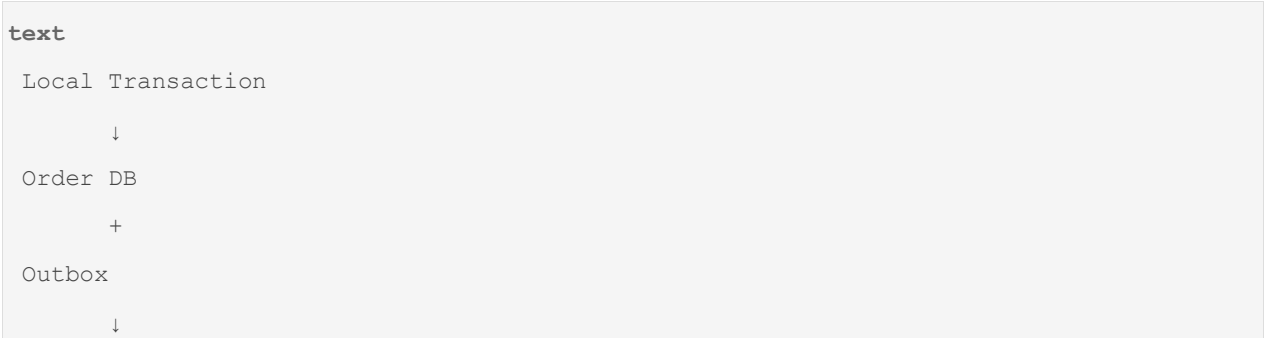


PAYMENT	INVENTORY	NOTIFICATION
SERVICE	SERVICE	SERVICE
Payment	Inventory	Notification
DB	DB	DB

And around the architecture:



And for consistency:



```
Kafka
  ↓
Other Services
  ↓
Eventual Consistency
  ↓
Saga / Compensation
```

That is the major conceptual jump from **Module 4 REST/API** to **Module 5 Microservices**.

The previous REST module itself ended by identifying timeout + bounded retry + exponential backoff + jitter + circuit breaker as the correct downstream-failure strategy, and then explicitly positioned distributed transaction/Saga design as the next Module 5 topic.

□ Highest-priority Module 5 questions for a 2-year interview

If the interviewer gives you only 20–30 minutes, prioritize:

Q1 Microservices **Q2** Monolith vs Microservices **Q4** Service boundaries **Q8** Database-per-Service **Q16** API Gateway **Q18** Service Discovery **Q22** Load Balancing **Q31** Network call vs method call **Q32** Timeout **Q35** Retry + duplicate request **Q36–38** Backoff + jitter + retry storm **Q39** Circuit Breaker **Q40** Cascading failure **Q46** Distributed transaction **Q47** Why `@Transactional` isn't enough **Q48** Saga **Q50** Choreography vs Orchestration **Q51** Eventual consistency **Q54** Idempotency **Q56** Payment succeeds but DB update fails **Q57** Outbox **Q59** Idempotent Consumer **Q61** Event-driven architecture **Q67–69** Observability/tracing/correlation ID **Q71** Order → Payment → Inventory **Q73** Downstream failure **Q74** Production incident **Q75** Complete system design

This also keeps continuity with the earlier material rather than duplicating it: the original interview bible explicitly intended later questions to move from definitions into **scenario-based reasoning, failure handling, scaling and design decisions** rather than memorized annotations.

Next in the sequence is Module 6 — 50 Spring Security/JWT questions, where we can go substantially deeper into the security-filter chain, authentication providers, JWT validation, OAuth2/OIDC, refresh tokens, stateless security, CSRF, CORS, method security, RBAC/ABAC, password encoding, session fixation, security exceptions and production security scenarios.